

Steve Yeh

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Placement Directors

Professor Mark Dean mark.dean@columbia.edu
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Education

Ph.D. in Economics, Columbia University. 2021–2027 (expected)
Committee: Evan Sadler (advisor), Navin Kartik, Qingmin Liu.
M.Phil in Economics, Columbia University. 2021–2024
M.A. in Economics, Columbia University. 2021–2023
B.A. in Economics and Mathematics, Cornell University 2017–2021
Summa Cum Laude in Economics and *Magna Cum Laude* in Mathematics.
Minors in Computer Science and Law & Society.

Research Fields

Networks, Microeconomic Theory.

Job Market Paper

[A Network Theory of Project Management](#)

Abstract: I study a principal who staffs projects by capping how much effort each worker can contribute to each project. Workers differ in which projects they prefer and work better alongside teammates, but find spreading effort across multiple projects costly. In an optimal assignment, a common ranking of projects divides each worker's schedule into three consecutive intervals: assigned without a binding cap, assigned but capped below what he would freely contribute, and unassigned. Project teams are necessarily nested. The principal tends to staff central workers to fewer projects because strong ties make a worker valuable to a team, but expensive to spread across teams. Whether a worker is a specialist or generalist is determined by which force prevails, captured by a ratio of two Katz-Bonacich centralities— one measuring his influence on his collaborators' total effort, the other his influence on their effort within a single project. Shocks propagate through the network induced by the optimal project assignment, and an improvement local to one project can degrade every other project in the organization.

Working Papers

Optimal Project Task Assignments

Revise & Resubmit at Theoretical Economics

Abstract: A principal assigns agents to tasks in a multi-task project. Agents tend to shirk at their tasks and rely on others' efforts because effort on one task substitutes for effort on similar tasks. Assigning the same agent to more tasks mitigates free-riding, but convex effort costs prevent him from completing all of his tasks. For a class of projects, an optimal task assignment assigns each agent to a module comprising a task central to the project and all locally related tasks. For general projects, modular task assignments are approximately optimal with a performance guarantee that depends on a project's task structure.

A Shocking Reversal: Flipping Complements and Substitutes in Network Games (with Evan Sadler)

Extended Abstract in the Proceedings of the 27th ACM Conference on Economics and Computation.

Abstract: In conventional network games, each player chooses one action that impacts all of her neighbors. We study players that select neighbor-specific efforts and face a convex total effort cost. Our main result highlights a striking reversal in comparative statics: When bilateral efforts are strategic complements (substitutes), shocks propagate through the network in a pattern befitting strategic substitutes (complements) in standard network games. This reversal, together with other fundamental changes in player behavior, suggests that retaining the single effort assumption when it is not appropriate may lead to poor policy recommendations.

Plausible Pairwise Stability

Abstract: I propose a stability concept for network formation under incomplete information. A network is plausibly pairwise stable if players' beliefs are consistent with common knowledge of network stability. I provide a general condition under which plausibly pairwise stable networks exhibit clustering. In a simple model of vertical differentiation, I characterize minimal plausibly pairwise stable networks. I show that incomplete information can stabilize segregated and sparse networks that are unstable under complete information. I discuss how incomplete information resolves the tension between stability and efficiency in the presence of negative linking externalities.

Referee Service

Program Committee for ACM Conference on Economics and Computation (2026), Journal of Economic Theory.

Presentations (* upcoming)

2026: Workshop on Advances in Economic Theory and Beyond (Tsinghua)*, Economics of Networks Workshop (Stony Brook), 37th Stony Brook International Conference on Game Theory, 27th ACM Conference on Economics and Computation (Rome), North American Summer Meeting of the Econometric Society (Emory), 11th Annual Conference on Network Science and Economics (University of Miami).

2025: 36th Stony Brook International Conference on Game Theory.

2024: Northwestern-Kellogg Summer School in Economic Theory, 35th Stony Brook International Conference on Game Theory.

Teaching

Intermediate Microeconomics (Undergraduate)	Spring 2023, Fall 2026
for Prof. Ingmar Nyman (2023) and Prof. Caterina Musatti (2026).	
Microeconomic Analysis II (MA)	Fall 2024
for Prof. Lilia Maliar.	
Principles of Economics (Undergraduate)	Fall 2023
for Prof. Waseem Noor.	
Microeconomic Analysis I (MA)	Fall 2022
for Prof. Bernard Salanié.	
Microeconomics I (MPA)	Summer 2023, 2024, 2025, 2026
for Prof. Muhammad Asali.	
Math Camp Instructor (MA, MPA, MIA)	Summer 2022, 2023, 2024, 2025

Awards

Dean's Fellowship, Columbia University	2021–Present
Wueller Fellowship for Best Teaching Fellow Runner-Up, Columbia University	2024
Harriss Award for Best 2nd-Year Paper Runner-Up, Columbia University	2023
Uri M. Possen Memorial Award for Best Undergraduate Honors Thesis, Cornell University	2021
Raymond & Bernice Conrad Endowed Scholarship, Cornell University	2019–2021
New York State Scholarship for Academic Excellence, Cornell University	2017–2021

Other Information

Employment: Goldman Sachs, Credit Risk Intern (Summer 2020).

Citizenship: US.

Languages: English (Native), Hakkanese (Native), Mandarin (Fluent), Spanish (Proficient).

References

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